
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Radiometric / Atmospheric Distortions and Corrections

Radiometric Preprocessing

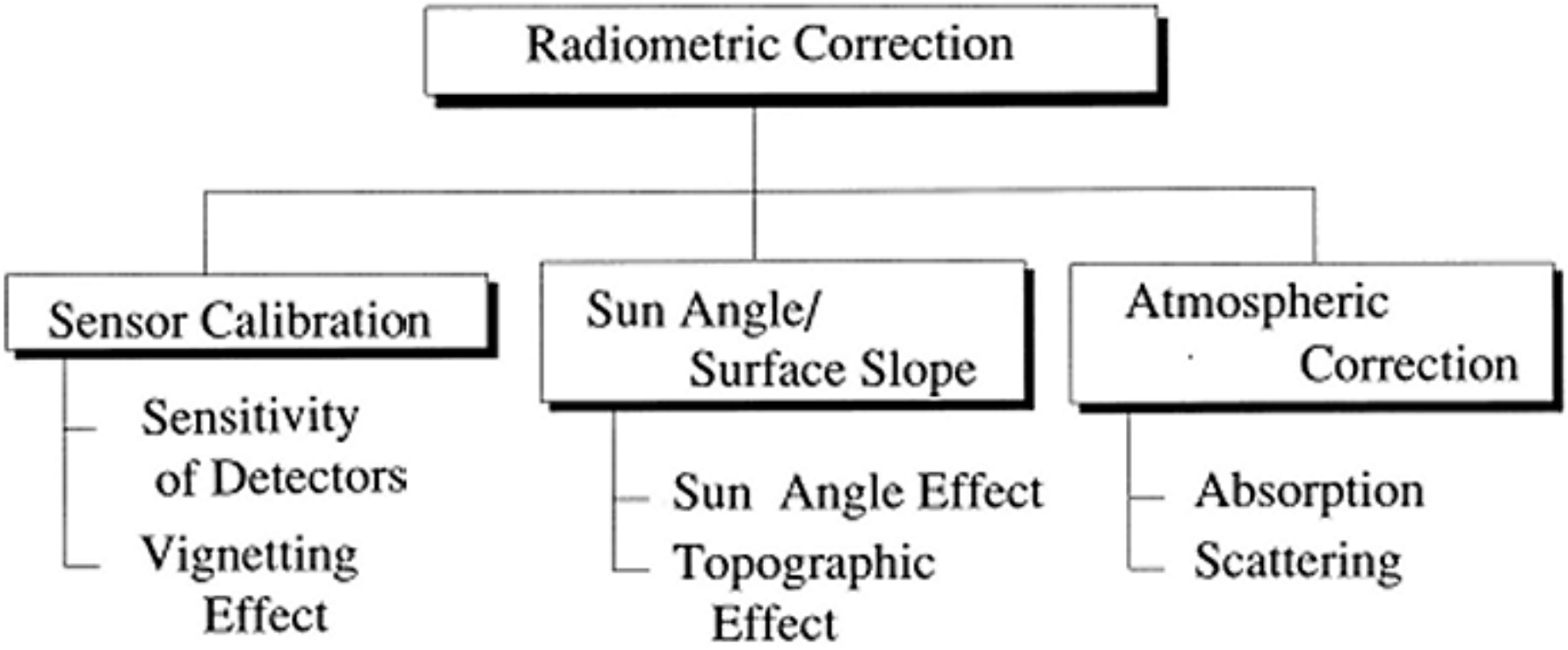
- **Radiometric preprocessing influences the brightness values of an image to correct for sensor malfunctions or to adjust the values to compensate for atmospheric degradation**

Radiometric Preprocessing

- **Need to perform correction for any or all of these influences depends directly upon the particular application at hand**
  - **To generate mosaics of images taken at different times or**
  - **To study the changes in the reflectance of ground features at different times**

Radiometric Corrections

- Radiometric corrections may be necessary due to variations in
  - Scene illumination and viewing geometry
  - Atmospheric conditions
  - Sensor noise and response
  - Topography
- Each of these will vary depending on
  - The specific sensor and platform used to acquire the data
  - The conditions during data acquisition


Radiometric Correction
Sensor Calibration
Sensitivity of Detectors
Vignetting Effect
Sun Angle/ Surface Slope
Sun Angle Effect
Topographic Effect
Atmospheric Correction
Absorption
Scattering

Radiometric Distortion- Sensor Sensitivity

- Radiometric distortions - Electro-Optical Type Sensor: Inaccurate representation of the radiance actually leaving the object of interest
  - Introduced by the sensor system itself when the **individual detectors do not function properly** or are **improperly calibrated**
    - **Missing scan line, bad pixels (shot noise)** covered in last semester

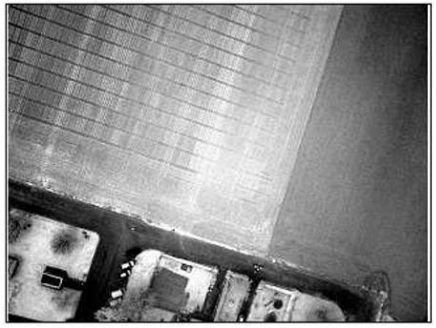
Radiometric Distortion- Sensor Sensitivity

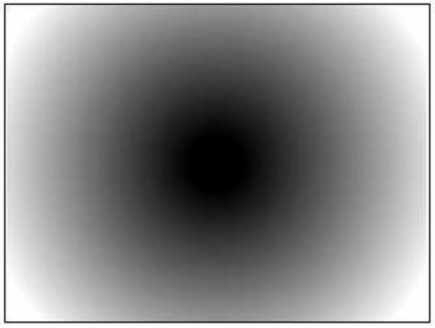
- **Lens Type Sensor**
  - **In the case of optical sensors, which use a lens, a fringe area in the corners will be darker as compared with the central area. This is called vignetting**
- **Correction: Using Anti vignetting Filter:**
  - **A glass plate made such that its transmissivity decrease from edge towards center**
  - **Darker in the central part and gradually becomes lighter in the peripheral region**

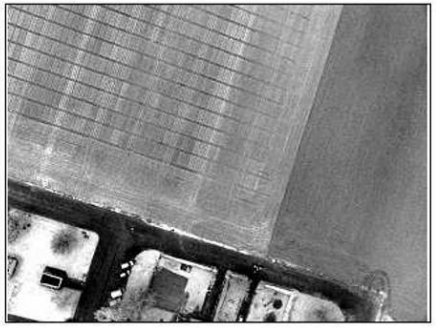
Anti Vignetting Filter









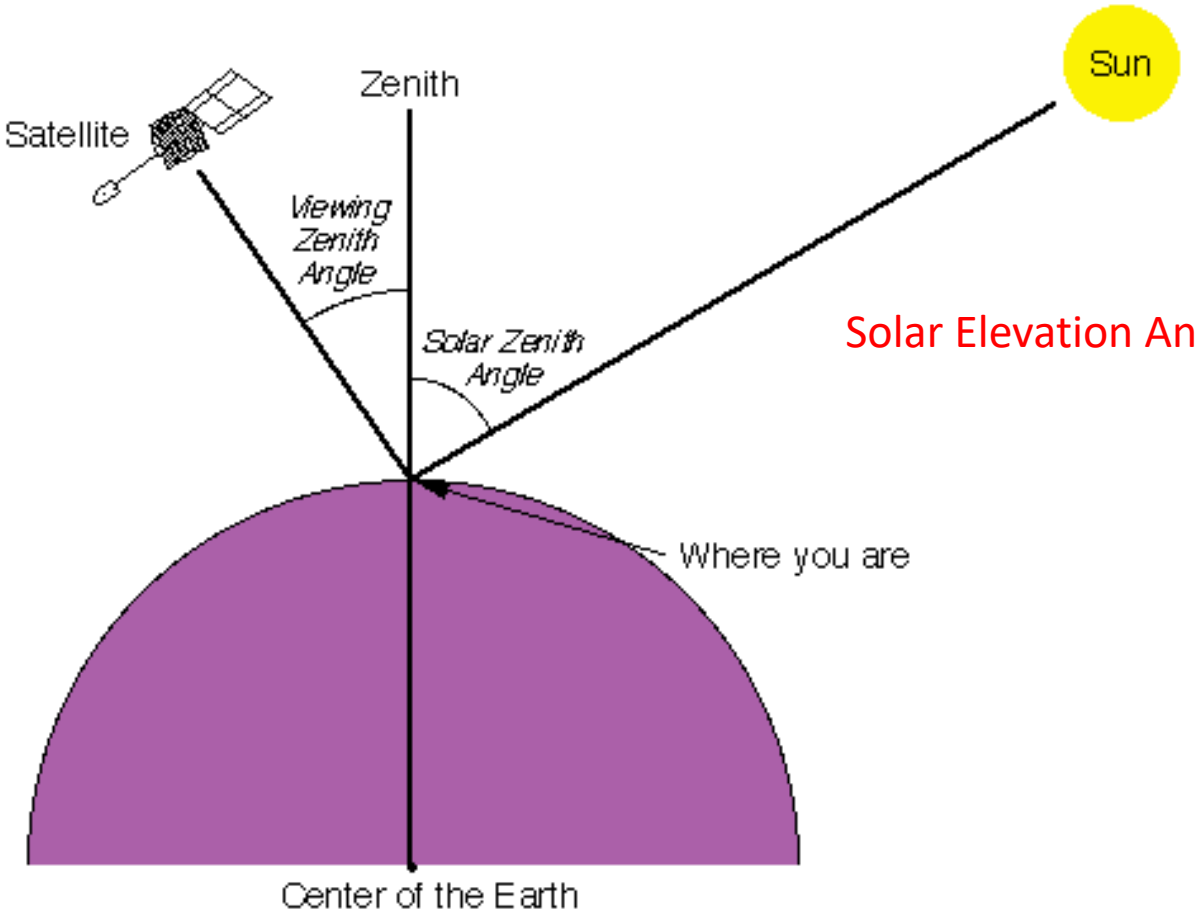


Illumination and View Angle Effects

- **Magnitude of the signal received at sensor depends upon:**
  - **Reflectance of the target**
  - **Effects of atmospheric interactions (nature and magnitude)**
  - **Slope and aspect of the target relative to the solar azimuth**
  - **Angle of view of sensor**
  - **Solar elevation/ zenith angle**

Solar Zenith Angle

Angle between the sun's rays and the normal to the Earth surface



Solar Elevation Angle?

Illumination and View Angle Effects

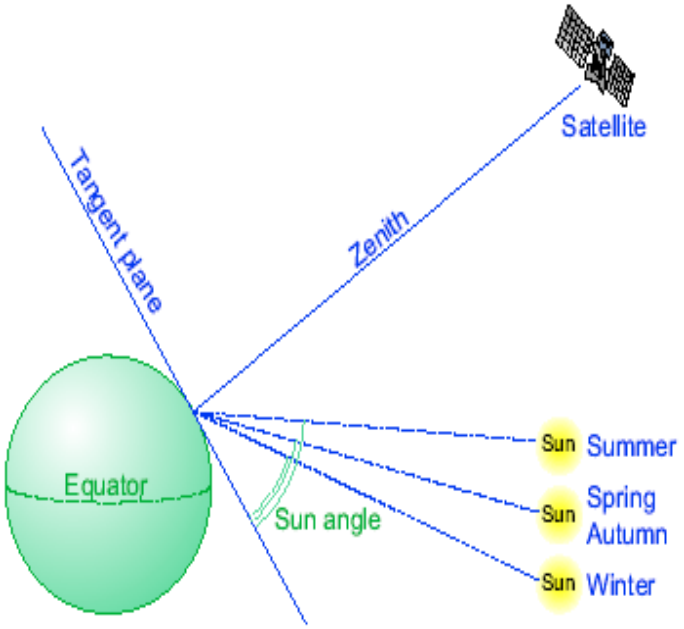
- **Upwelling radiance of a particular target has to pass through the atmospheric, therefore angle of view of sensor will affect the upwards path length**
- **If atmosphere is present, irradiance at the target will be reduced depending upon the atmospheric path length**
- **Atmospheric path length will increase as solar elevation decreases, and the degree of atmospheric interference will increase**
- **Nadir view will be less affected than off Nadir**

Illumination and View Angle Effects

- In the absence of atmosphere and no other complicating factors, the magnitude of the radiance recorded by the sensor depends upon:
  - Illumination angle
  - Angel of view of sensor
- The reflectance of target will vary as these two angles alter
- Slope and aspect angles of the target

Sun Angle Correction

- **The image data of different seasons are acquired under different solar illumination**
- **Sun angle correction becomes more important when one wants to generate mosaics taken at different times or perform change detection studies**



Effects of seasonal changes on the solar elevation angle in the northern hemisphere

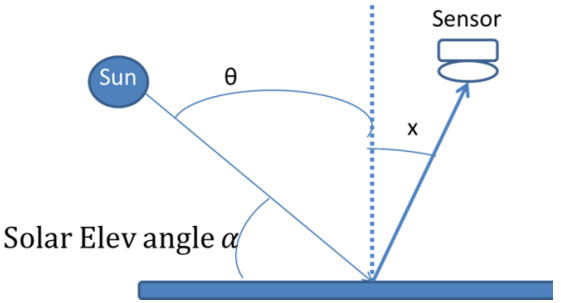
The sun elevation correction accounts for the seasonal position of the sun relative to the earth

Sun Elevation Correction

- **Through this process, image data acquired under different solar illumination angles are normalized by calculating pixel brightness values assuming the sun was at the zenith on each date of sensing**

Sun Elevation Correction

- The correction is usually applied by dividing each pixel value in a scene by the sine of the solar elevation angle (α) for the particular time



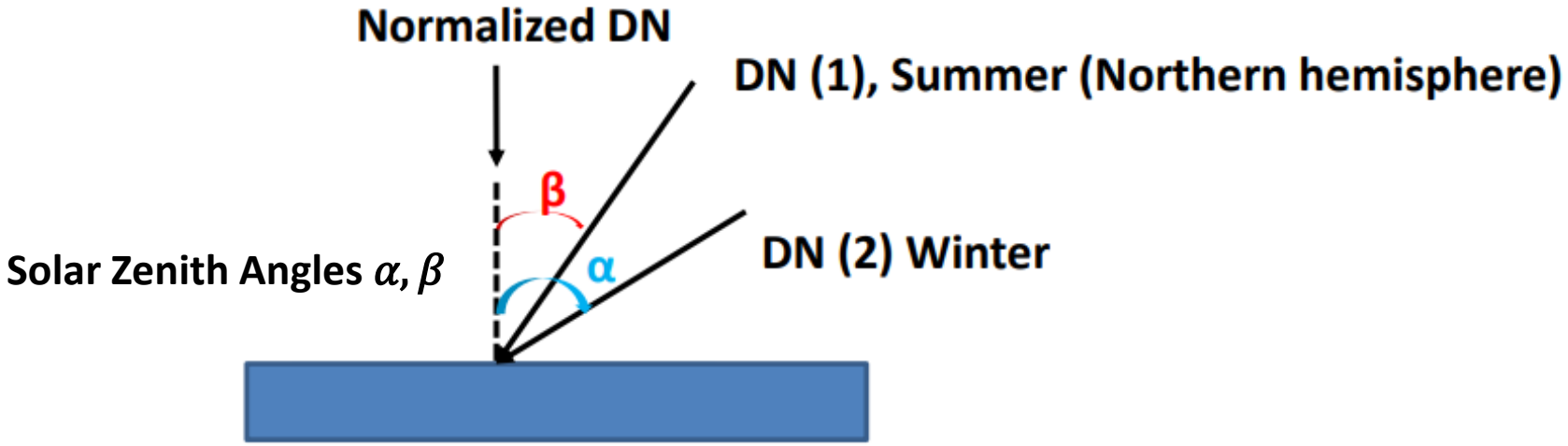
- Correction:

$$DN_c = DN_a/(Sin\alpha)$$

- DN_c = Sun angle corrected (Normalized DN)
- DN_a = Actual DN
- Alternatively, the correction is applied in terms of the sun's angle from the zenith, which is simply 90° minus the solar elevation angle
- In either case, the correction ignores topographic and atmospheric effects

Sun Elevation Correction

Alternatively, the correction is applied in terms of the sun's angle from the zenith, which is simply 90° minus the solar elevation angle



$$Normalized\ DN(1) = \frac{DN(1)}{Cos\beta}$$

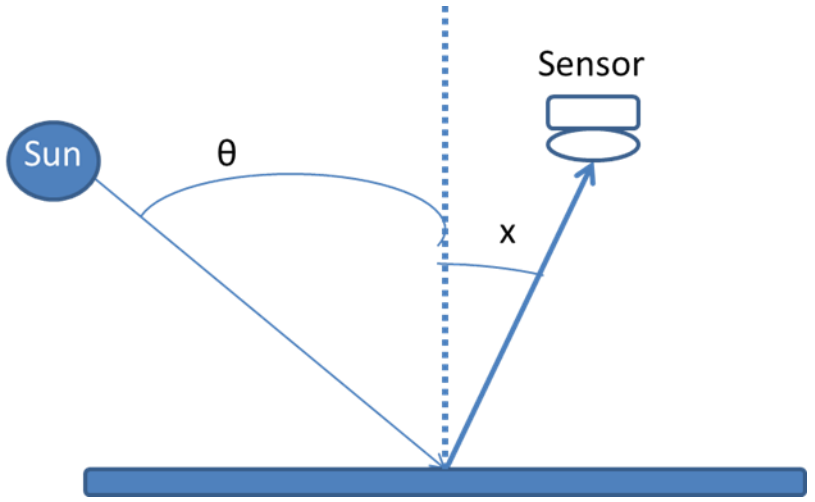
$$Normalized\ DN\ (2) = \frac{DN(2)}{Cos\alpha}$$

Radiometric Correction for View Angle Effects

- If the solar zenith angle is θ, Observed radiance is *L*, Desired view angle is *X*, then the correction is simply,

$$L' = L\frac{\cos x}{\cos \theta}$$

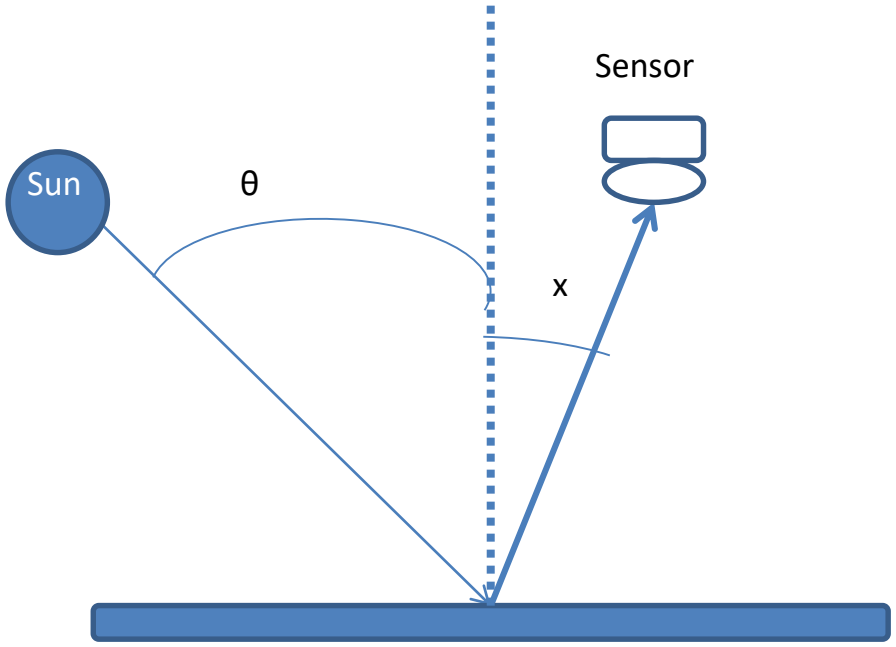
- It may be used to standardize temporal images to a standard solar illumination angle



- Sensor elevation angle will also affect the upward path length

Illumination and View Angle Effects

- **Amounts of shadowing will also be dependent upon the solar elevation angle**



Radiometric Distortions-Topography

- Digital imagery from mountainous regions, often contains radiometric distortion known as topographic effects
- Topographic effects result from the differences in illumination due to the angle of the sun and the angle of the terrain
- This causes a variation in image brightness values
- Different BVs for same object on sunlit and shadowed side

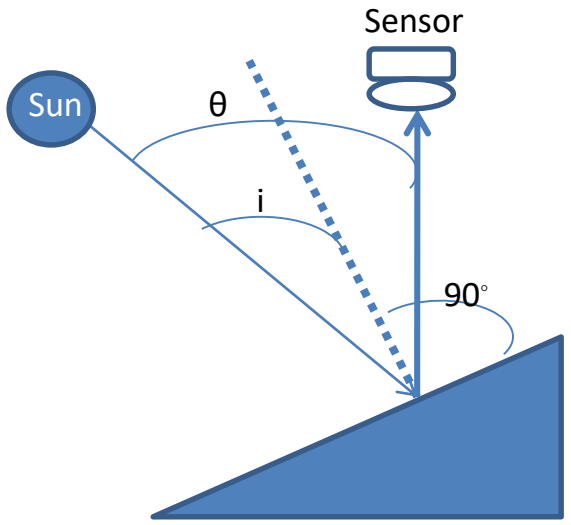
Radiometric Distortions-Topography

- **Similar land covers – different reflectance's!**
- **Results in variable illumination conditions (in extreme cases data are lost due to excessive shadowing)**
- **Difference in radiance values from inclined surfaces compared to horizontal ones**

Topographic Slope- Aspect Correction
(Cosine Correction)

- **The amount of the irradiance reaching a pixel on slope is directly proportional to the cosine of the incidence angle (i) , the angle between normal to the pixel and the sun**
- **Only the part cos(i) of the total incoming irradiance reaches the inclined pixel**

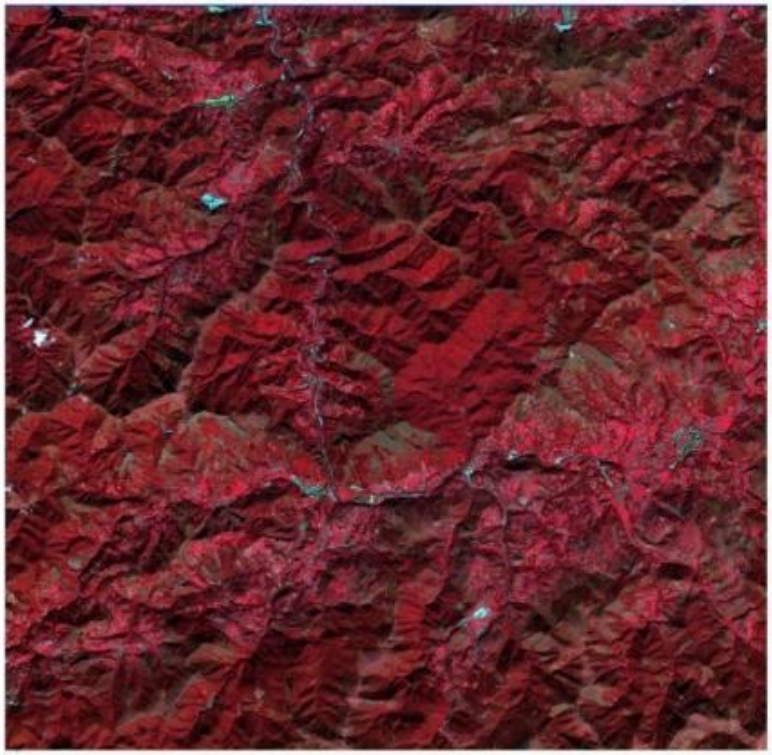
$$L' = L\frac{\cos(\theta)}{\cos(i)}$$



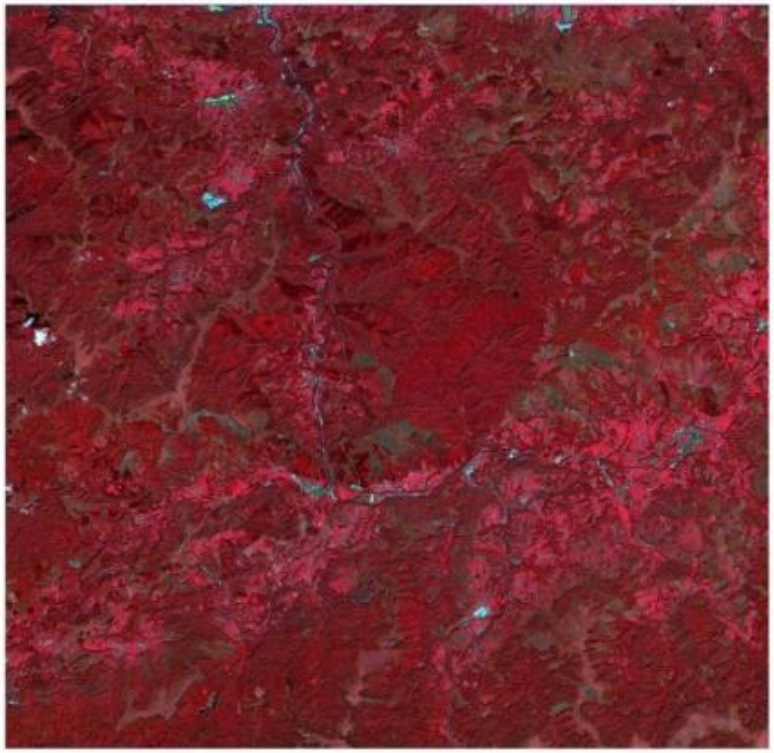
where: L'= radiance observed for a horizontal surface (i.e., slope-aspect corrected remote sensor data)
L= radiance observed over sloped terrain (i.e., the raw remote sensor data)

It only cater for direct irradiance illuminating the pixel

Radiometric Distortions-Topography



Before



After

?